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CHAPTER 1 • NUMBER • TOPIC 4

Powers & Standard Form

Big and tiny numbers written as $a \times 10^n$.
Move the decimal point, multiply the numbers and add the powers — and let the calculator check.



$$\frac{3}{10} = 0.3 =$$

30%

6

VARIANTS

45

PAST-PAPER PARTS

53

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Powers & roots

V2 Standard form → ordinary

V3 Ordinary → standard form

V4 $\times \div$ and powers

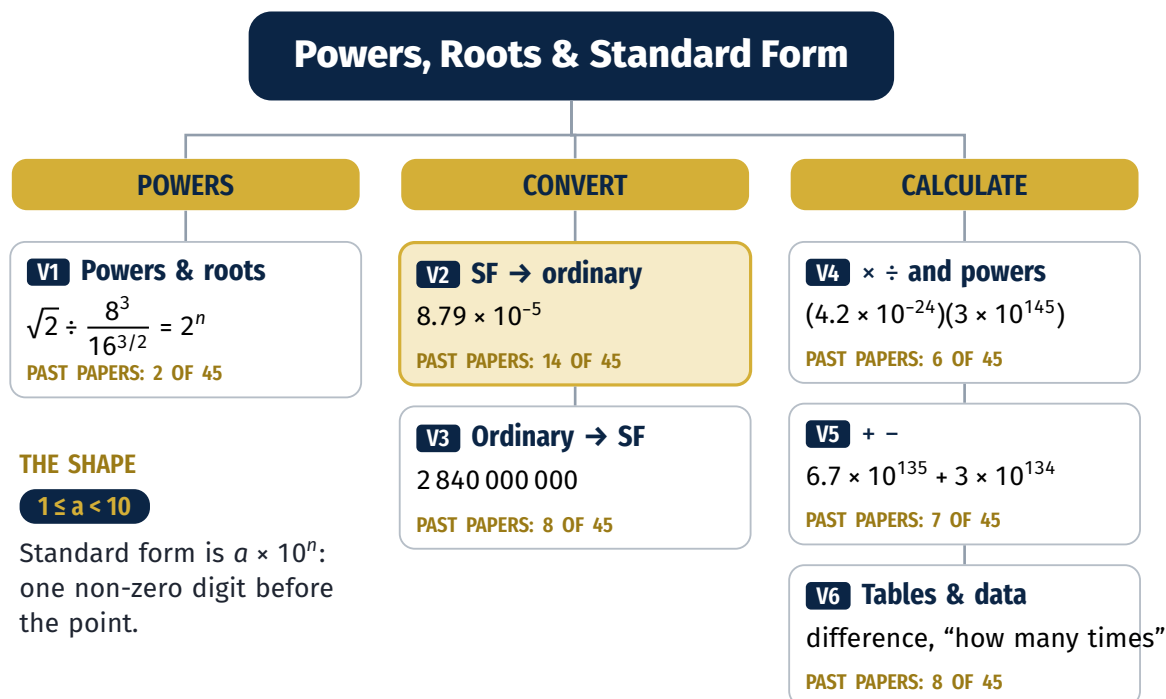
V5 $+$ $-$ in standard form

V6 Tables & real data

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Powers & Standard Form at a glance



Your notes


How often does each variant come up?

24 past-paper questions (Jan 2020 → Nov 2025) hold 45 parts; 53 practice parts — grouped by type.

	PAST PAPERS (OUT OF 45)	PRACTICE PARTS
V1 Powers & roots	2	20
V2 SF → ordinary	14	8
V3 Ordinary → SF	8	9
V4 × ÷ and powers	6	10
V5 + - in SF	7	2
V6 Tables & data	8	4

WHAT THIS TELLS YOU A standard-form question is on almost every Paper 1: two 1-mark conversions (V2, V3) and one calculation (V4/V5). That is 3–4 marks to bank.

BEFORE YOU START

The big idea



Standard form: $a \times 10^n$, with
 $1 \leq a < 10$.

The power n counts how many places
the decimal point **moves**.

Big number $\rightarrow$ positive power.

Small number (less than 1) $\rightarrow$
negative power.

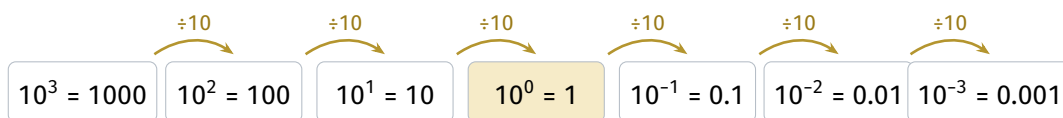
$$5\,200\,000 = 5.2 \times 10^6$$

$$0.000\,087 = 8.7 \times 10^{-5}$$

a BETWEEN 1 AND 10 • n = PLACES MOVED



Powers of 10 and the power ladder

**Zero power**

$a^0 = 1$
anything⁰ (except
0)

Negative

$a^{-n} = \frac{1}{a^n}$
flip

Fraction

$a^{\frac{m}{n}} = (\sqrt[n]{a})^m$
root, then power

Laws

$\times$ add $\div$ subtract
same base

**CALCULATOR FIRST, THEN CHECK**

Type standard form with $\times 10^x$: $4.2 \times 10^x (-) 24$. For huge powers like 10^{145} the calculator gives **Math ERROR** — then do numbers and powers separately by hand (v4), and use the calculator only for the numbers.

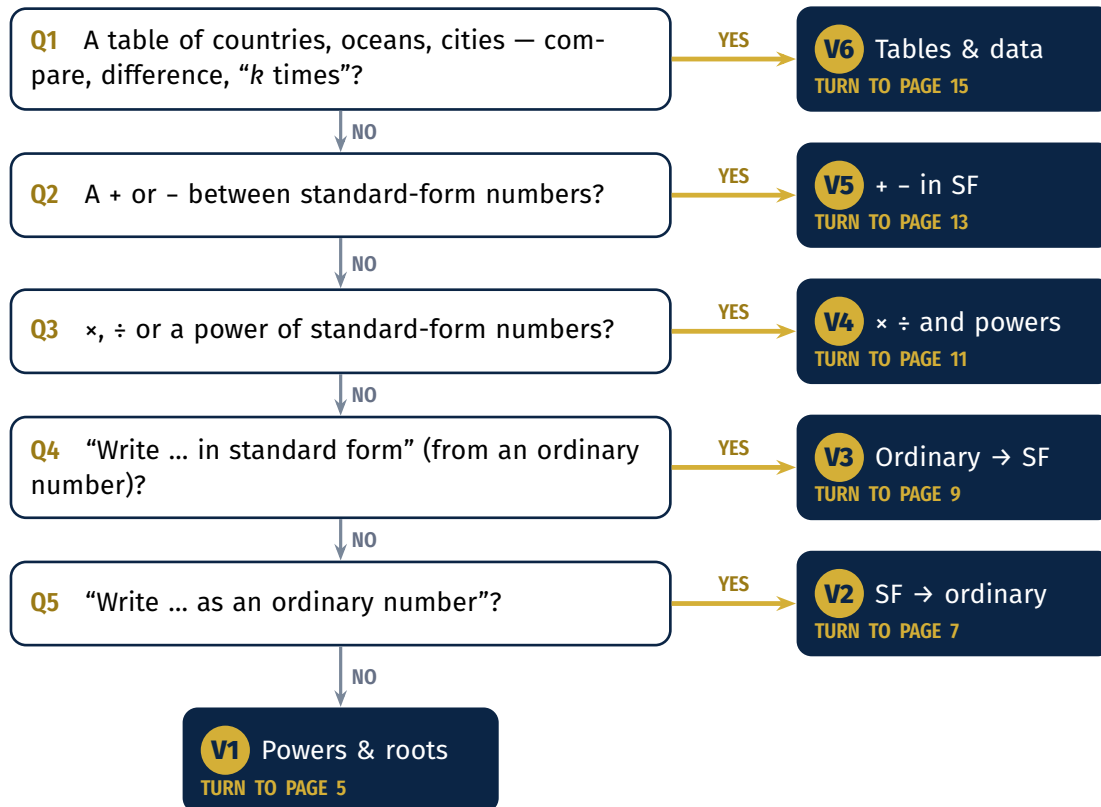
COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.

**COLOUR CODE IN THE WORKING**

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1

Powers & Roots of Numbers

Zero, negative and fractional powers — and “find n ” with the same base.

SEEN IN PAST PAPERS

2 / 45

PRACTICE QUESTIONS: 20

 Your notes

Came up in: Jan 2021 P2H Q6(a) · May 2023 P1H Q15(a)

? What is it?

In plain words

Zero power → 1. **Negative power** → flip ($2^{-3} = \frac{1}{8}$). **Fraction power** → bottom is the root, top is the power ($27^{\frac{2}{3}} = 3^2 = 9$).

“= 2^n ”: write everything as a **power of 2**, then use the laws.

IT LOOKS LIKE

5^0

3^{-2}

$27^{\frac{2}{3}}$

$64^{\frac{1}{4}}$

$2^2 = 4$

$2^1 = 2$

$2^0 = 1$

$2^{-1} = \frac{1}{2}$

$2^{-2} = \frac{1}{4}$

$2^{-3} = \frac{1}{8}$

EACH STEP $\div 2$

How do I do it?

1 Zero / negative

$a^0 = 1$; a^{-n} : flip to $\frac{1}{a^n}$.

2 Fraction power

Root (bottom) first, then power (top).

3 Same base

To find n : write every number as a power of one prime ($8 = 2^3$, $\sqrt{2} = 2^{\frac{1}{2}}$).

4 Laws

$\times$ add, $\div$ subtract, then match the powers.

NEGATIVE = FLIP · FRACTION = ROOT

Full detail: Algebra Topic 2.

V1 Powers & Roots of Numbers

VARIANT 1 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Write down the value of y^0 ” → **1**

“= 2^n . Work out the value of n ” → **Everything as powers of 2, then match**

“Show clear working” with roots → **Write each power of 2 on its own line**

**WORKED EXAMPLE****PAST PAPER**

May 2023 • Paper 1H • Q15(a) • 3 marks

$\sqrt{2} \div \frac{8^3}{16^{\frac{3}{2}}} = 2^n$. Work out the value of n . Show your working clearly.

STEP 1 $8^3 = (2^3)^3 = 2^9$, $16^{\frac{3}{2}} = (2^4)^{\frac{3}{2}} = 2^6$ ◀ every number as a power of 2

STEP 2 $\frac{2^9}{2^6} = 2^3$, $\sqrt{2} = 2^{\frac{1}{2}}$ ◀ divide: subtract

STEP 3 $2^{\frac{1}{2}} \div 2^3 = 2^{\frac{1}{2}-3} = 2^{-\frac{5}{2}}$ ◀ divide again

STEP 4 $n = -\frac{5}{2}$ ◀ match the powers

ANSWER $n = -\frac{5}{2}$

✓ $2^{-2.5} = 0.1768 = \sqrt{2} \div 8$ ✓

**WATCH OUT**

✗ $3^{-2} = -9$

✓ $3^{-2} = \frac{1}{9}$

A negative power flips — it never makes the number negative.

**EXAM TIP**

Check on the calculator: both sides of the equation give the same decimal.

**Check yourself**

calculator first • answers on page 17

1.1

Jan 2021 P2H

Write down the value of y^0 .

Answer:

1.2

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Work out $27^{\frac{2}{3}}$.

Answer:

1.3

Practice • Easy

Work out 3^{-2} .

Answer:

V2**VARIANT 2** MOST COMMON IN THE EXAM**Standard Form → Ordinary Number**Move the decimal point n places: right for $+n$, left for $-n$.

SEEN IN PAST PAPERS

14 / 45

PRACTICE QUESTIONS: 8



Your notes

Came up in: Jan 2020 P1H Q6(a) · Oct/Nov 2020 P1H Q5(a) · Jan 2021 P1H Q6(b) · Jan 2022 P1HR Q8(a) · Jun 2022 P1H Q8(b) · Jun 2022 P2HR Q9(a) · Jun 2023 P2H Q6(b) · May 2023 P1HR Q8(a) · Nov 2023 P1H Q9(a) · May 2024 P1H Q9(a) · Jun 2024 P2HR Q9(a) · Nov 2024 P1H Q9(a) · Jun 2025 P1H Q8(b) · Nov 2025 P2H Q8(a)

? What is it?**In plain words** $\times 10^5$: move the point**5 places right** (the number gets bigger). $\times 10^{-5}$: move it **5 places left**

(smaller than 1). Fill the gaps with zeros.

IT LOOKS LIKE

8.79×10^{-5}

6.04×10^5

7.8×10^{-4}

5 PLACES LEFT



0 · 0 0 0 0 8 7 9

$8.79 \times 10^{-5} = 0.000\,087\,9$

How do I do it?**1 Read the power**

Positive → bigger (right). Negative → smaller (left).

2 Move the pointExactly n places.**3 Fill with zeros**

Every jumped empty place is a 0.

4 Check size 10^{-5} → four zeros after the point before the first digit.**+ RIGHT · - LEFT** Count the jumps.

V2 Standard Form → Ordinary Number

VARIANT 2 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Write 8.79×10^{-5} as an ordinary number”

→ **Point 5 places left**

“Write 6.04×10^5 as an ordinary number”

→ **Point 5 places right: 604 000**

(1 mark)

→ **Count carefully — there is no method mark****WORKED EXAMPLE** **PAST PAPER**

June 2025 · Paper 1H · Q8(b) · 1 mark

Write 8.79×10^{-5} as an ordinary number.**STEP 1** power -5: smaller than 1, move left ◀ read the power**STEP 2** $8.79 \rightarrow 0.879 \rightarrow 0.0879 \rightarrow \dots$ five jumps ◀ move the point**STEP 3** 0.000 087 9 ◀ fill with zeros**ANSWER** 0.000 087 9✓ calculator: $8.79 \times 10^{-5} = 0.0000879$ **WATCH OUT**

✗ 0.000 008 79

✓ 0.000 087 9

10^{-5} moves 5 places: four zeros after the point, then 8.

EXAM TIP

Type the standard form on the calculator and press **S⇌D** or switch to Norm mode to see the ordinary number.

Check yourself

calculator first · answers on page 17

2.1 Jan 2020 P1HWrite 7.8×10^{-4} as an ordinary number.

Answer:

2.2 Jun 2024 P2HRWrite 6.04×10^5 as an ordinary number.

Answer:

2.3 Nov 2023 P1HWrite 5.87×10^{-4} as an ordinary number.

Answer:

V3

VARIANT 3

Ordinary Number → Standard Form

Put the point after the first non-zero digit;
count the jumps for the power.

SEEN IN PAST PAPERS

8 / 45

PRACTICE QUESTIONS: 9



Your notes

Came up in: Oct/Nov 2020 P1H Q5(b) • Jan 2021 P1H Q6(a) • Jan 2022 P1HR Q8(b) • Jun 2022 P1H Q8(a) • Jun 2023 P2H Q6(a) • Nov 2023 P1H Q9(b) • Jun 2024 P2HR Q9(b) • Jun 2025 P1H Q8(a)

? What is it?

In plain words

Find the **first non-zero digit**; the number a starts there (2.84).

Count how many places the point moved to get there: that is n — **positive** for a big number, **negative** for a small one. Drop the end zeros.

$$2.840000000$$

9 PLACES

$$2.84 \times 10^9$$

IT LOOKS LIKE

2 840 000 000

0.000 089

520 million

How do I do it?

1 First digit

The first non-zero digit — the point goes right after it.

2 Count

How many places from there to the real point?

3 Sign

Big number $+n$; number less than 1 $-n$.

4 Write

$a \times 10^n$ with $1 \leq a < 10$, no end zeros.

$1 \leq a < 10$ 28.4×10^8 is not standard form.

V3 Ordinary Number → Standard Form

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"Write 2 840 000 000 in standard form" → 2.84, **9 jumps**: 2.84×10^9

"Write 0.000 089 in standard form" → 8.9, **5 jumps left**: $\times 10^{-5}$

"520 million" → $520\,000\,000 = 5.2 \times 10^8$

**WORKED EXAMPLE** **PAST PAPER**

January 2021 • Paper 1H • Q6(a) • 1 mark

Write 2 840 000 000 in standard form.

STEP 1

2.84

◀ first non-zero digit, point after it

STEP 2

from 2.84 to 2 840 000 000 the point moves 9 places

◀ count

STEP 3

a big number → +9

◀ sign

ANSWER 2.84×10^9 ✓ calculator: $2.84 \times 10^9 = 2\,840\,000\,000$ **WATCH OUT**✗ 284×10^7 ✓ 2.84×10^9

a must be between 1 and 10 — one digit before the point.

**EXAM TIP**

"Million" is 10^6 : 520 million
 $= 520 \times 10^6 = 5.2 \times 10^8$.

**Check yourself**

calculator first • answers on page 17

3.1

Jun 2022 P1H

Write 0.000 089 in standard form.

Answer:

3.2

Jun 2023 P2H

Write 76 000 000 in standard form.

Answer:

3.3

Jun 2025 P1H

Write 520 million in standard form.

Answer:

V4

VARIANT 4

Multiply, Divide & Powers

Numbers with numbers, powers of 10 with powers of 10 — then fix a back to $1 \leq a < 10$.

SEEN IN PAST PAPERS

6 / 45

PRACTICE QUESTIONS: 10



Your notes

Came up in: Jan 2022 P1H Q8 • Jan 2022 P2H Q12 • Jan 2022 P1HR Q8(c) • May 2023 P1H Q15(b) • Nov 2024 P1H Q9(b) • Jun 2025 P1H Q8(c)

? What is it?

In plain words

Split it: multiply (or divide) the **numbers**, and use the index laws on the **powers of 10**.

If the number part is not between 1 and 10, **fix it**:

$$12.6 \times 10^{121} = 1.26 \times 10^{122}.$$

IT LOOKS LIKE

$$(4.2 \times 10^{-24})(3 \times 10^{145}) \quad a^3$$

$$\frac{4 \times 10^{512}}{1.6 \times 10^{300}}$$

$$(4.2 \times 10^{-24}) \times (3 \times 10^{145})$$

$$= 12.6 \times 10^{121}$$

FIX: $\div 10$, POWER +1

$$= 1.26 \times 10^{122}$$

How do I do it?

1 Split

Numbers together, powers of 10 together.

2 Numbers

Multiply / divide them (calculator).

3 Powers $\times$ add, $\div$ subtract, $()^3$ multiply by 3.**4 Fix**Number ≥ 10 : $\div 10$, power +1. Number < 1 : $\times 10$, power -1.**NUMBERS $\times$ NUMBERS • POWERS ADD** Then fix a .

V4 Multiply, Divide & Powers

VARIANT 4 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Work out $a \times b$. Give your answer in standard form”

→ **Numbers $\times$, powers add, fix**

“ $a = 6 \times 10^{40}$. Work out a^3 ”

→ $6^3 = 216$, **power $\times 3$: fix to 2.16**

Huge powers (calculator error)

→ **Do the powers by hand****WORKED EXAMPLE****PAST PAPER**

January 2022 · Paper 1H · Q8 · 2 marks

$a = 4.2 \times 10^{-24}$, $b = 3 \times 10^{145}$. Work out $a \times b$. Give your answer in standard form.

STEP 1 $(4.2 \times 3) \times (10^{-24} \times 10^{145})$ ◀ split

STEP 2 12.6 ◀ numbers

STEP 3 $10^{-24+145} = 10^{121}$ ◀ powers add

STEP 4 $12.6 \times 10^{121} = 1.26 \times 10^{122}$ ◀ fix: 12.6 is not < 10

ANSWER 1.26×10^{122}

✓ 1.26 is between 1 and 10 ✓

**WATCH OUT**

✗ 12.6×10^{121} (left as it is)

✓ 1.26×10^{122}

Not standard form until $1 \leq a < 10$.

**EXAM TIP**

When you divide a by 10, the power goes **up** by 1 — the value stays the same.

**Check yourself**

calculator first · answers on page 17

4.1

Jan 2022 P2H

$a = 6 \times 10^{40}$. Work out a^3 in standard form.

Answer:

4.2

Nov 2024 P1H

Work out $(6.5 \times 10^{-40}) \times (8 \times 10^{185})$ in standard form.

Answer:

4.3

Jun 2025 P1H

Work out $(5 \times 10^{42}) \times (7 \times 10^{-180})$ in standard form.

Answer:

V5

VARIANT 5

Add & Subtract in Standard Form

You can only add like powers: make the powers of 10 the same first (or use ordinary numbers).

SEEN IN PAST PAPERS

7 / 45

PRACTICE QUESTIONS: 2



Your notes

Came up in: Jan 2020 P1H Q6(b) • Oct/Nov 2020 P1H Q5(c) • Jan 2021 P2H Q6(b) • May 2023 P1HR Q8(b) • Jun 2024 P2HR Q9(c) • Jun 2025 P1HR Q14 • Nov 2025 P2H Q8(b)

? What is it?**In plain words**

$6.7 \times 10^{135} + 3 \times 10^{134}$: the powers differ, so rewrite one:

$3 \times 10^{134} = 0.3 \times 10^{135}$. Now add the numbers: 7×10^{135} .

With small powers, just change to **ordinary numbers** (or use the calculator).

IT LOOKS LIKE

$$\frac{5.6 \times 10^4 + 7 \times 10^3}{2.8 \times 10^{-3}}$$

$$6.7 \times 10^{135} + 3 \times 10^{134}$$

$$\begin{array}{r} 6.7 \times 10^{135} \\ + 0.3 \times 10^{135} \quad \text{SAME POWER FIRST} \\ \hline 7.0 \times 10^{135} \end{array}$$

How do I do it?**1 Same power**

Rewrite the smaller one with the bigger power
($3 \times 10^{134} = 0.3 \times 10^{135}$).

2 Add / subtract

Add the numbers, keep the power.

3 Small powers

Or: change to ordinary numbers and use the calculator.

4 Fix

Answer in standard form, $1 \leq a < 10$.

SAME POWER BEFORE + OR - Never add the powers here.

V5 Add & Subtract in Standard Form

VARIANT 5 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

$“6.7 \times 10^{135} + 3 \times 10^{134}”$ → **Same power first:** 0.3×10^{135}

A fraction with + or – on top → **Top first (ordinary numbers), then divide**

“Give your answer in standard form” → **Fix a at the end**

WORKED EXAMPLE **PAST PAPER**

June 2025 · Paper 1HR · Q14 · 2 marks

Work out $6.7 \times 10^{135} + 3 \times 10^{134}$. Give your answer in standard form.

STEP 1 $3 \times 10^{134} = 0.3 \times 10^{135}$ ◀ same power first

STEP 2 $6.7 \times 10^{135} + 0.3 \times 10^{135} = 7.0 \times 10^{135}$ ◀ add the numbers

ANSWER 7×10^{135}

✓ $1 \leq 7 < 10$ ✓**WATCH OUT**

× 9.7×10^{269} (added numbers and powers)

✓ 7×10^{135}

Adding numbers does not add powers — that rule is for multiplying.

**EXAM TIP**

For $\frac{5.6 \times 10^4 + 7 \times 10^3}{2.8 \times 10^{-3}}$: type it on the calculator with brackets round the top.

**Check yourself**

calculator first · answers on page 17

5.1

Jan 2020 P1H

Work out
 $\frac{5.6 \times 10^4 + 7 \times 10^3}{2.8 \times 10^{-3}}$ in
 standard form.

Answer:

5.2

Oct/Nov 2020 P1H

Work out
 $\frac{3 \times 10^5 - 2.7 \times 10^4}{6 \times 10^{-2}}$ in
 standard form.

Answer:

5.3

Nov 2025 P2H

Work out
 $\frac{3 \times 10^5 + 8 \times 10^3}{4 \times 10^{-2}}$ in
 standard form.

Answer:

V6

VARIANT 6

Tables & Real Data

Compare, subtract or divide real quantities given in standard form.

SEEN IN PAST PAPERS

8 / 45

PRACTICE QUESTIONS: 4



Your notes

Came up in: Jan 2020 P2HR Q10 • Jan 2021 P2HR Q8 • May 2021 P1H Q8 • Nov 2021 P2H Q9 • Jan 2022 P2HR Q10 • Jun 2022 P2HR Q9 • Nov 2023 P1H Q9(c) • May 2024 P1H Q9(b)

? What is it?

In plain words

Biggest / smallest: compare the **powers first**, then the numbers.

Difference: subtract (V5). “*k* times” or “per person”: **divide**.

Round only at the end, to what is asked.

IT LOOKS LIKE

difference

“Pacific is *k* times Arctic”

Arctic	4.539×10^4
Atlantic	1.119×10^5
Pacific	1.357×10^5
Southern	1.797×10^4

BIGGEST POWER WINS

How do I do it?

1 Read the table

Which rows does the question use?

2 Compare

Bigger power → bigger number; same power → compare the numbers.

3 Calculate

Difference: subtract. “Times” / per person: divide.

4 Round & form

Standard form or the rounding asked (2 s.f., nearest whole).

POWERS FIRST WHEN COMPARING 10^5 beats 10^4 whatever the numbers.

V6 Tables & Real Data

VARIANT 6 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Which has the greatest ...?” → **Biggest power, then biggest number**

“Work out the difference ...” → **Subtract; answer in standard form**

“... is k times ...” / “per person” → **Divide**

WORKED EXAMPLE **PAST PAPER**

January 2021 · Paper 2HR · Q8 · 3 marks

Coastlines: Arctic 4.539×10^4 , Atlantic 1.119×10^5 , Pacific 1.357×10^5 , Indian 6.653×10^4 , Southern 1.797×10^4 km. (a) Which is greatest? (b) Atlantic – Southern, in standard form.

STEP 2 10^5 : Atlantic and Pacific; $1.357 > 1.119$ ◀ (a) powers first

STEP 3 $111\,900 - 17\,970 = 93\,930$ ◀ (b) subtract

STEP 4 9.393×10^4 ◀ standard form

ANSWER (a) Pacific (b) 9.393×10^4 km ✓ calculator agrees

WATCH OUT

✗ “Southern is biggest — $1.797 > 1.357$ ”

✓ compare the powers first

1.797×10^4 is 17 970 — much smaller than 135 700.

EXAM TIP

Subtract on the calculator using $\times 10^x$ — no need to write the ordinary numbers.

Check yourself

calculator first · answers on page 17

6.1 May 2021 P1H

China 1.4×10^9 ,
Germany 8.2×10^7 . Find
the difference in
standard form.

Answer:

6.2 Jun 2022 P2HR

Tokyo 3.7×10^7 ,
Ahmedabad 7.7×10^6 .
Difference, standard
form, 2 s.f.?

Answer:

6.3 Nov 2023 P1H

Human brain 8.5×10^{10}
neurons, monkey
 1.47×10^9 . Human = $K \times$
monkey. Find K (1 d.p.).

Answer:

CHECK YOURSELF

Answers

Every answer below was checked with sympy (check_answers.py).

1.1 1

1.2 9

1.3 $\frac{1}{9}$

2.1 0.000 78

2.2 604 000

2.3 0.000 587

3.1 8.9×10^{-5}

3.2 7.6×10^7

3.3 5.2×10^8

4.1 2.16×10^{122}

4.2 5.2×10^{146}

4.3 3.5×10^{-137}

5.1 2.25×10^7

5.2 4.55×10^6

5.3 7.7×10^6

6.1 1.318×10^9

6.2 2.9×10^7

6.3 $K = 57.8$

THE WHOLE TOPIC ON ONE PAGE

Powers & Standard Form cheat sheet



Your notes
V1 Powers & roots 2/45

$$27^{\frac{2}{3}} = 9, 3^{-2} = \frac{1}{9}$$

Negative = flip, fraction = root.

V2 SF → ordinary 14/45

$$8.79 \times 10^{-5} = 0.0000879$$

+ right, - left.

V3 Ordinary → SF 8/45

$$2\,840\,000\,000 = 2.84 \times 10^9$$

 $1 \leq a < 10$; count the jumps.**V4 $\times \div$ 6/45**

$$12.6 \times 10^{121} = 1.26 \times 10^{122}$$

Numbers $\times$, powers add, fix.**V5 + - 7/45**

$$6.7 \times 10^{135} + 0.3 \times 10^{135}$$

Same power first.

V6 Tables 8/45

$$1.119 \times 10^5 - 1.797 \times 10^4$$

Powers first; subtract or divide.

**TOP 3 LOST MARKS**

1. 12.6×10^{121} not fixed $\times$
2. one zero too many / too few $\times$
3. adding the powers in a sum $\times$

BOTTOM LINE $a \times 10^n$ with $1 \leq a < 10$ · $+n$ big, $-n$ small · $\times$: powers add · $+$: same power first.